Harvard University
Computer Science 20

Test 5
Final Exam

Student Name: ________________________________

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This is an individual, closed-book, closed-note assessment. You may not collaborate with anyone else or use any external resources during the test.

PROBLEM 1

For sets A and B, define the *symmetric difference* as $A \triangle B = (A \setminus B) \cup (B \setminus A)$.

(A) Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$. (Hint: show $\subseteq$ in both directions.)

(B) Using part (a), prove the cardinality formula $|A \triangle B| = |A| + |B| - 2|A \cap B|$ for finite sets A and B. You may assume the inclusion–exclusion principle $|A \cup B| = |A| + |B| - |A \cap B|$ without proof.

(C) State a condition on A and B that characterizes exactly when $A \triangle B = A \cup B$. (No justification required.)

Solution.

(A) ($\subseteq$) Let $x \in A \triangle B = (A \setminus B) \cup (B \setminus A)$. Then either $x \in A$ and $x \notin B$, or $x \in B$ and $x \notin A$. In either case $x \in A \cup B$ and $x \notin A \cap B$, so $x \in (A \cup B) \setminus (A \cap B)$.
($\supseteq$) Let $x \in (A \cup B) \setminus (A \cap B)$. Then $x \in A \cup B$ and $x \notin A \cap B$. Since $x \in A$ or $x \in B$ but x is not in both, x is in exactly one of them, so $x \in (A \setminus B) \cup (B \setminus A) = A \triangle B$. ■

(B) By part (a), $A \triangle B = (A \cup B) \setminus (A \cap B)$. Since $A \cap B \subseteq A \cup B$, the elements of $A \triangle B$ are exactly the elements of $A \cup B$ that are not in $A \cap B$, so

$$|A \triangle B| = |A \cup B| - |A \cap B|.$$

By inclusion–exclusion, $|A \cup B| = |A| + |B| - |A \cap B|$. Substituting,

$$|A \triangle B| = |A| + |B| - |A \cap B| - |A \cap B| = |A| + |B| - 2|A \cap B|. \quad ■$$

(C) $A \triangle B = A \cup B$ iff A and B are disjoint (i.e., $A \cap B = \emptyset$).

PROBLEM 2

Consider the claim: *For all positive integers a and b, if ab is a perfect square, then both a and b are perfect squares.* (Recall: a positive integer m is a *perfect square* if there exists a positive integer j with $m = j^2$.)

(A) Disprove the claim by giving an explicit counterexample with concrete values of a and b. Verify that your counterexample satisfies the hypothesis but not the conclusion.

(B) Prove the *strengthened claim:* For all positive integers a and b with $\gcd(a, b) = 1$, if ab is a perfect square, then both a and b are perfect squares. You may use without proof the fact that every positive integer has a unique prime factorization.

Solution.

(A) Take $a = 2$ and $b = 8$. Then $ab = 16 = 4^2$ is a perfect square. But $a = 2$ is not a perfect square (since $1^2 = 1$ and $2^2 = 4$ skip over 2), and $b = 8$ is not a perfect square (since $2^2 = 4$ and $3^2 = 9$ skip over 8). The hypothesis holds but the conclusion fails.

(B) If $ab = k^2$, then by unique prime factorization every prime p appears in ab with even total exponent. Now decompose a and b into their prime factorizations. The hypothesis $\gcd(a, b) = 1$ means no prime appears in both a and b, so each prime's contribution to the exponent in ab comes entirely from a alone or from b alone. Since the total exponent of each prime is even and that exponent lives entirely in one factor, the exponent in a alone (and likewise in b alone) must already be even. Hence a and b are each perfect squares. The counterexample in part (a) violates this hypothesis: $\gcd(2, 8) = 2 \neq 1$, so the prime 2 appears in both a and b and the exponents can "cancel" to even in the product without being even in each factor individually.

PROBLEM 3

Let R be a relation on a set A with $|A| \geq 2$. Recall that irreflexivity means that for every $x \in A$, $(x, x) \notin R$.

(A) Prove the following: *If R is symmetric, transitive, and irreflexive, then $R = \emptyset$ (i.e., no pair is related).*

(B) Show that the conclusion of part (a) can fail if the transitive hypothesis is dropped. Give a concrete non-empty relation on $A = \{1, 2, 3\}$ that is symmetric and irreflexive but not transitive, and verify your example.

Solution.

(A) Suppose for contradiction that $R \neq \emptyset$. Then there exist $x, y \in A$ with $(x, y) \in R$. By symmetry, $(y, x) \in R$. By transitivity applied to (x, y) and (y, x), $(x, x) \in R$. But R is irreflexive, so $(x, x) \notin R$. Contradiction. Hence $R = \emptyset$. ■

(B) Take $R = \{(1, 2), (2, 1)\}$. Symmetric: yes ($(1, 2)$ and $(2, 1)$ are both present). Irreflexive: no (x, x) pair. Transitive fails: $(1, 2), (2, 1) \in R$ but $(1, 1) \notin R$. R is non-empty.

PROBLEM 4

On the Cartesian plane, a *lattice path* is a sequence of unit steps either right $(+1, 0)$ or up $(0, +1)$. For example, $(0, 0) \to (1, 0) \to (1, 1) \to (2, 1)$ is a lattice path from $(0, 0)$ to $(2, 1)$.

Consider lattice paths from the origin $(0, 0)$ to the point $(5, 3)$.

(A) How many such paths exist with no further restrictions?

(B) How many such paths pass through the point $(3, 2)$?

Solution.

(A) Each path uses 5 rights and 3 ups for a total of 8 steps. Choose which 3 are "up":

$$\binom{8}{3} = 56.$$

(B) A path through $(3, 2)$ is the concatenation of a path from $(0, 0)$ to $(3, 2)$ (which uses 3 rights and 2 ups, $\binom{5}{2} = 10$ paths) with a path from $(3, 2)$ to $(5, 3)$ (which uses 2 rights and 1 up, $\binom{3}{1} = 3$ paths). By the product rule, the count is $10 \cdot 3 = 30$.

PROBLEM 5

(A) Verify that $7 \mid 8^n - 1$ for $n = 1, 2, 3$ by direct computation.

(B) Prove by induction that $7 \mid 8^n - 1$ for every integer $n \geq 1$.

Solution.

(A)

- $n = 1$: $8^1 - 1 = 7$, divisible by 7.
- $n = 2$: $8^2 - 1 = 63 = 7 \cdot 9$, divisible by 7.
- $n = 3$: $8^3 - 1 = 511 = 7 \cdot 73$, divisible by 7.

(B) Let $P(n) := (7 \mid 8^n - 1)$. Proof for all $n \geq 1$ by induction.
Base case ($n = 1$). $8 - 1 = 7$ and $7 \mid 7$, so $P(1)$ holds.
Inductive step. Assume $P(k)$ holds: $7 \mid 8^k - 1$, so $8^k - 1 = 7m$ for some integer m, i.e., $8^k = 7m + 1$. Show $P(k+1)$: $7 \mid 8^{k+1} - 1$.

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(7m + 1) - 1 = 56m + 8 - 1 = 56m + 7 = 7(8m + 1).$$

Since $8m + 1 \in \mathbb{Z}$, $7 \mid 8^{k+1} - 1$. Hence $P(k+1)$ holds, and by induction $P(n)$ holds for all $n \geq 1$. ■

PROBLEM 6

Let S be the smallest set of integer pairs satisfying:

- **Base:** $(0,0) \in S$.
- **C1:** If $(x,y) \in S$, then $(x+1,\ y+x+1) \in S$.

(A) Show that $(3,6) \in S$ by exhibiting an explicit derivation. List each rule application and the resulting pair.

(B) Prove by structural induction that every $(x,y) \in S$ satisfies $y = \dfrac{x(x+1)}{2}$.

Solution.

(A) Derivation of $(3,6)$:

1. Base: $(0,0) \in S$.
2. C1: $(0+1, 0+0+1) = (1,1) \in S$.
3. C1: $(1+1, 1+1+1) = (2,3) \in S$.
4. C1: $(2+1, 3+2+1) = (3,6) \in S$. ✓

Sanity check: $\dfrac{3 \cdot 4}{2} = 6$. ✓

(B) **Base case.** $(x,y) = (0,0)$: $\dfrac{0 \cdot 1}{2} = 0 = y$. ✓

Inductive step. Assume $(x,y) \in S$ has $y = \dfrac{x(x+1)}{2}$. Then C1 produces $(x', y') = (x+1, y+x+1)$. Compute:

$$y' = y + x + 1 = \frac{x(x+1)}{2} + (x+1) = \frac{x(x+1) + 2(x+1)}{2} = \frac{(x+1)(x+2)}{2} = \frac{x'(x'+1)}{2}.$$

By structural induction, every $(x,y) \in S$ satisfies $y = \dfrac{x(x+1)}{2}$. ∎

PROBLEM 7

A disease has prevalence $\pi = 0.1$. There are two independent tests:

- Test A: true-positive rate 0.9, false-positive rate 0.1.
- Test B: true-positive rate 0.8, false-positive rate 0.2.

Conditional on the true disease status, the two test results are independent. A patient is given both tests.

(A) Compute $P(\text{disease} \mid A^+ \cap B^+)$, the probability of having the disease given both tests are positive.

(B) Compute $P(\text{disease} \mid A^+ \cap B^-)$, the probability of having the disease when test A is positive but test B is negative.

Solution.
Let $D =$ disease ($P(D) = 0.1$, $P(\overline{D}) = 0.9$).

(A)

$$P(A^+B^+ \mid D) = 0.9 \cdot 0.8 = 0.72, \quad P(A^+B^+ \mid \overline{D}) = 0.1 \cdot 0.2 = 0.02.$$

$$P(D \mid A^+B^+) = \frac{0.72 \cdot 0.1}{0.72 \cdot 0.1 + 0.02 \cdot 0.9} = \frac{0.072}{0.072 + 0.018} = \frac{0.072}{0.090} = \frac{4}{5} = 0.8.$$

(B)

$$P(A^+B^- \mid D) = 0.9 \cdot 0.2 = 0.18, \quad P(A^+B^- \mid \overline{D}) = 0.1 \cdot 0.8 = 0.08.$$

$$P(D \mid A^+B^-) = \frac{0.18 \cdot 0.1}{0.18 \cdot 0.1 + 0.08 \cdot 0.9} = \frac{0.018}{0.018 + 0.072} = \frac{0.018}{0.090} = \frac{1}{5} = 0.2.$$

PROBLEM 8

You roll a fair six-sided die repeatedly until you see either a 1 or a 2 (whichever comes first). Let N be the number of rolls needed. Since the number of rolls is a geometric random variable with success probability $1/3$, $\mathbb{E}[N] = 3$. You may use this fact without proof.

(A) Let T be the value of the *final* roll (so $T \in \{1, 2\}$). Compute $\mathbb{E}[T]$.

(B) Let S be the sum of the values of all rolls (including the final roll that produced the 1 or 2). Compute $\mathbb{E}[S]$.

Solution.

(A) By symmetry, at each roll a 1 and a 2 are equally likely (each occurs with probability $1/6$ independently of past rolls). Hence the first of $\{1, 2\}$ to appear is equally likely to be either value, so

$$\mathbb{E}[T] = \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot 2 = \tfrac{3}{2} = 1.5.$$

(B) Decompose $S =$ (sum of the first $N - 1$ rolls) $+$ (value of the final roll). The first $N - 1$ rolls are each conditionally uniform on $\{3, 4, 5, 6\}$ (since they did not equal 1 or 2), with conditional mean

$$\frac{3 + 4 + 5 + 6}{4} = \frac{18}{4} = 4.5.$$

By linearity of expectation, using $\mathbb{E}[N] = 3$,

$$\mathbb{E}[\text{sum of first } N - 1] = \mathbb{E}[N - 1] \cdot 4.5 = 2 \cdot 4.5 = 9.$$

The final roll has $\mathbb{E}[T] = 1.5$ from part (a). Therefore

$$\mathbb{E}[S] = 9 + 1.5 = 10.5.$$

PROBLEM 9

A small library has four events scheduled this week. Each event needs a meeting room. Two events conflict (cannot share a room) iff their time windows overlap. (Shared endpoints do not count as overlap: e.g., an event from 9:00–10:30 does not conflict with an event from 10:30–12:00.)

The events $E_1, \ldots, E_4$ have the following time windows:

- E_1: 9:00–10:30
- E_2: 10:00–11:30
- E_3: 10:00–11:00
- E_4: 11:00–12:00

(A) Build the conflict graph G (the graph with one vertex per event and an edge between two vertices iff the events conflict).

(B) Find the chromatic number $\chi(G)$. Justify both the lower and upper bounds.

Solution.

(A) Two windows conflict iff they overlap (share more than just an endpoint). The edges are:

$$E(G) = \{\{E_1, E_2\},\ \{E_1, E_3\},\ \{E_2, E_3\},\ \{E_2, E_4\}\}.$$

(Check: E_1 and E_4 do not overlap since E_1 ends at 10:30 and E_4 starts at 11:00. E_3 and E_4 touch at 11:00 only, so they do not conflict.)

(B) $\chi(G) = 3$.
Lower bound: $\{E_1, E_2, E_3\}$ form a triangle: E_1–E_2 (overlap 10:00–10:30), E_1–E_3 (overlap 10:00–10:30), E_2–E_3 (overlap 10:00–11:00). A triangle requires at least 3 colors, so $\chi(G) \geq 3$.
Upper bound: The 3-coloring

$$E_1 \mapsto 1, \quad E_2 \mapsto 2, \quad E_3 \mapsto 3, \quad E_4 \mapsto 1$$

is proper: each of the 4 edges has endpoints with different colors (verify on each edge from part (a)). Hence $\chi(G) \leq 3$.

PROBLEM 10

Student Tom proposes a new, simplified algorithm for finding a minimum-weight spanning tree (MST) on a connected weighted graph G with n vertices: *For $n-1$ steps, add the edge of minimum weight that has not already been added.*

(A) Draw a connected weighted graph on 4 vertices with 5 edges on which Tom's algorithm successfully produces an MST. Specify the edge weights and report the total MST weight.

(B) Draw a connected weighted graph on 4 vertices on which Tom's algorithm fails — that is, the set of $n-1=3$ edges output by Tom's algorithm is *not* a spanning tree. Specify the weights and explain in 1–2 sentences why the output is not a spanning tree.

(C) Suggest a one-line modification to Tom's algorithm that would make it correct (i.e., always produce an MST on connected weighted graphs). Which well-known MST algorithm does it become after your modification?

Solution.

(A) Take $V=\{A,B,C,D\}$ and edges with weights:

$$AB=1,\quad BC=2,\quad CD=3,\quad AD=10,\quad AC=20.$$

Sorted by weight: $AB(1), BC(2), CD(3), AD(10), AC(20)$. Tom adds AB, BC, CD — total weight 6. The result is a path $A-B-C-D$, which is a spanning tree of weight 6. This is the MST (any other spanning tree must include an edge of weight ≥ 10).

(B) Take $V=\{A,B,C,D\}$ and edges with weights:

$$AB=1,\quad AC=2,\quad BC=3,\quad AD=100,\quad BD=101.$$

Sorted: $AB(1), AC(2), BC(3), AD(100), BD(101)$. Tom adds the three lightest edges: AB, AC, BC. But these three edges form a triangle on $\{A,B,C\}$ and do not include vertex D. The output has 3 edges, 3 vertices used, and 1 vertex isolated — not a spanning tree (a spanning tree must touch all 4 vertices and be acyclic).

(C) *Modified algorithm:* For $n-1$ steps, add the edge of minimum weight that has not already been added *and that does not create a cycle with the edges already added*. This is exactly **Kruskal's algorithm**. The cycle-avoidance check forces the partial edge set to remain a forest, and after $n-1$ added edges this forest has $n-1$ edges on n vertices — so it is a spanning tree by the edge-count characterization of trees.

Scratch paper.

Scratch paper.